Synergy or Star Power: Optimizing Esports Team Rosters

Duong Vu

August 27, 2026

1 Introduction

I have always found the game League of Legends fascinating. As one of the most popular multiplayer online battle arena (MOBA) games worldwide, it features two teams of five players competing in strategic, fast-paced matches. Recently, I have been paying more attention to the esports aspect of this game. As in any other team sport, roster construction during the transfer season is a highly debated topic among fans, analysts, and team organizations. During these periods, considerable attention is paid to high-profile player signings, and there is often strong speculation about how changes in individual talent will translate into team success. However, teams with individually top-rated players do not always outperform more balanced rosters in actual competition.

Building on these observations, I wanted to understand how team synergy impacts success beyond individual talent. To this end, I will compare two roster-construction strategies: a greedy approach that selects the best-rated player for each role independently, and a synergy-aware approach that incorporates pairwise player compatibility. This comparison quantitatively assesses the extent to which overall performance is driven by team synergy versus individual skills, and the extent to which individual skills can be traded for team synergy. Ultimately, this project aims to provide practical insights for players, coaches, and organizations trying to build a more effective team.

2 Motivation and Relevance

The problem of roster building is becoming increasingly relevant across esports in general, and in League of Legends in particular, as the stakes are rising rapidly. The total yearly prize pool for League of Legends esports grew from 9 million USD in 2024 to over 14 million USD in 2025 [1]. Additionally, esports offers an unusually rich data environment: nearly every match is logged in detail, creating large, high-resolution datasets well-suited for quantitative analysis. As the financial and competitive stakes continue to grow, data-driven approaches to team-building become increasingly essential.

Furthermore, while this study focuses on League of Legends, the core question of how to balance individual talent versus team compatibility is broadly relevant to all team-based sports. In competitive sports, acquiring the highest-performing players often requires a significant financial investment, as they are highly sought after. Developing a framework to evaluate the tradeoff between individual skill and team synergy can help organizations build stronger teams and potentially reduce unnecessary costs.

3 Data Sources and Description

This study uses publicly available professional League of Legends match data, primarily sourced from Oracle's Elixir [4]. The dataset includes detailed per-match statistics, including player performance metrics, match outcomes, laning statistics, objective control, champion selections, and historical teammate information. These data provide both individual-level features for evaluating player skill and relational information for estimating player synergy.

For practical reasons, the analysis is restricted to matches from three recent seasons (2023–2025) in the League of Legends Champions Korea (LCK). This region was chosen for its large volume of available match data and the consistency of its competitive structure. Limiting the study to a single region also avoids confounding effects introduced by differences in playstyles, regional policies, and tournament formats. Additionally, only players who competed in the LCK across all three seasons are included, ensuring each player has a sufficiently long and consistent match history to support estimation of both individual performance and teammate-level interactions.

4 Methodology

4.1 Optimization Model

We start by formulating roster construction as an optimization problem. A League of Legends team consists of five distinct roles: top, jungle, mid, bot, and support. We index these roles by $r = 1, \ldots, 5$. Each team must select exactly one player for each role. Let $\mathcal{P}_r$ denote the set of available players for role r, and let $n_r = |\mathcal{P}_r|$ denote the number of such players.

We define a binary decision variable $x_{ir} \in \{0, 1\}$ for $1 \le i \le n_r$, where $x_{ir} = 1$ indicates that player i is selected for role r. Let q_{ir} denote the quality of player i in role r, and let $s_{irjr'}$ denote the synergy between player i in role r and player j in role r'.

Our objective is to maximize the total team score, defined as the sum of individual player quality and pairwise synergy terms:

$$\sum_{r=1}^{5} \sum_{i=1}^{n_r} q_{ir} x_{ir} + \sum_{r=1}^{5} \sum_{r'=r+1}^{5} \sum_{i=1}^{n_r} \sum_{j=1}^{n_{r'}} s_{irjr'} x_{ir} x_{jr'}. \tag{1}$$

This objective is subject to the constraint that exactly one player is selected for each role:

$$\sum_{i=1}^{n_r} x_{ir} = 1 \quad \forall r = 1, \ldots, 5 \tag{2}$$

Written in this form, the objective contains quadratic terms of the form $x_{ir} x_{jr'}$. To obtain a computationally efficient linear formulation, we introduce auxiliary binary variables $y_{irjr'} \in \{0, 1\}$, which indicate whether player i in role r and player j in role r' are selected together. The objective then becomes

$$\sum_{r=1}^{5} \sum_{i=1}^{n_r} q_{ir} x_{ir} + \sum_{r=1}^{5} \sum_{r'=r+1}^{5} \sum_{i=1}^{n_r} \sum_{j=1}^{n_{r'}} s_{irjr'} y_{irjr'}, \tag{3}$$

We enforce the logical relationship between x and y using the following constraints:

- $y_{irjr'}$ can only be 1 if player i for role r is selected:

$$y_{ijrr'} \leq x_{ir} \quad \forall i, j,\ r < r'. \tag{4}$$

- $y_{irjr'}$ can only be 1 if player j for role r' is selected:

$$y_{irjr'} \leq x_{jr'} \quad \forall i, j,\ r < r'. \tag{5}$$

- $y_{irjr'}$ must be 1 if both players are selected:

$$y_{irjr'} \geq x_{ir} + x_{jr'} - 1 \quad \forall i, j,\ r < r'. \tag{6}$$

Combining everything together, our optimization problem is as follows

$$\begin{aligned}
\max\ & \sum_{r=1}^{5}\sum_{i=1}^{n_r} q_{ir}x_{ir} + \sum_{r=1}^{5}\sum_{r'=r+1}^{5}\sum_{i=1}^{n_r}\sum_{j=1}^{n_{r'}} s_{irjr'}y_{irjr'} \\
\text{subjected to } & \sum_{i=1}^{n_r} x_{ir} = 1 \quad \forall r = 1, \ldots, 5, \\
& y_{irjr'} \leq x_{ir} \qquad \forall i = 1, \ldots, n_r, \forall j = 1, \ldots, n_{r'},\ \forall r < r' \\
& y_{irjr'} \leq x_{jr'} \qquad \forall i = 1, \ldots, n_r, \forall j = 1, \ldots, n_{r'},\ \forall r < r' \\
& y_{irjr'} \geq x_{ir} + x_{jr'} - 1 \quad \forall i = 1, \ldots, n_r, \forall j = 1, \ldots, n_{r'},\ \forall r < r' \\
& x_{ir} \in \{0, 1\} \qquad \forall i = 1, \ldots, n_r, \forall r = 1, \ldots, 5 \\
& y_{irjr'} \in \{0, 1\} \quad \forall i = 1, \ldots, n_r, \forall j = 1, \ldots, n_{r'},\ \forall r < r'
\end{aligned} \tag{7}$$

4.2 Estimating Player Quality

After formulating the optimization model, the main challenge is estimating player quality and player synergy. The dataset contains per-match statistics and outcomes for each player. To translate these statistics into a measurable notion of player impact, we use a predictive model that estimates match win probability from performance features. Since the statistical profiles of different roles differ substantially, we train a separate predictive model for each role.

For each role, we use a logistic regression classifier to predict the probability of winning a match from a vector of in-game statistics $\boldsymbol{x}_r$. The model takes the form

$$p = \sigma(\boldsymbol{\beta}_r^T \boldsymbol{x}_r + \beta_r^0). \tag{8}$$

where $\sigma(\cdot)$ is the sigmoid function, given by

$$\sigma(x) = \frac{1}{1 + e^{-x}}. \tag{9}$$

For each player i in role r, we compute their average statistics across all matches, denoted $\bar{\boldsymbol{x}}_{ir}$, and define their raw quality score as

$$q_{ir}^{\text{raw}} = \boldsymbol{\beta}_r^T \bar{\boldsymbol{x}}_{ir}. \tag{10}$$

To obtain comparable scores within each role, we standardize these raw scores across all players in the same role:

$$q_{ir} = \frac{q_{ir}^{\text{raw}} - \text{mean}(q_r^{\text{raw}})}{\text{std}(q_r^{\text{raw}})}. \tag{11}$$

After computing these scores, the highest-ranked players for each role aligned well with widely recognized top performers in the LCK (see, for example [2, 3]). Table 1 shows the top three players for each role along with their normalized quality scores.

Top		Jungle		Mid		Bot		Support	
name	score	name	score	name	score	name	score	name	score
Kiin	1.437	Canyon	1.054	Chovy	1.611	Gumayusi	0.988	Delight	1.304
Doran	1.174	Peanut	0.468	Faker	0.714	Viper	0.885	Keria	0.770
Zeus	0.582	Oner	-0.256	Zeka	0.508	Aiming	0.848	Lehends	0.735

Table 1: Top three estimated players by normalized quality score for each role.

4.3 Estimating Player Synergy

Estimating synergy between players is substantially more challenging and prone to bias. Strong teams tend to recruit strong players, and players often remain on the same roster for extended periods, leading to highly correlated performance and sparse observations across many player pairs. As a result, many pairs of players have never played together.

After experimenting with multiple estimators, we adopt an empirical win-rate-based measure. For each pair of players, we compute their historical win rate when playing together. For player pairs with no shared match history, we assign a neutral win rate of 0.5. The synergy score is defined as this win rate shifted by 0.5 so that it is centered at zero. That is,

$$s_{irjr'} = \begin{cases} \text{empirical win rate} - 0.5 & \text{if the pair has played together,} \\ 0 & \text{otherwise.} \end{cases} \tag{12}$$

5 Results

After running the optimization model, we obtained an optimal roster that balances both individual player quality and team synergy. For comparison, we also constructed a "greedy" roster by selecting the highest-rated player in each role, ignoring interaction effects among players.

Table 2 shows the two rosters and their corresponding normalized quality scores. The two agree at three of the five positions: both take Kiin, Canyon, and Chovy, and they disagree only on the bot lane and support. Although the greedy roster achieves a higher total individual quality score (6.393 vs. 5.685), the optimized roster more than makes up the difference in synergy. It sacrifices 0.708 points of total individual quality (6.393 − 5.685) in exchange for 1.067 additional points of synergy (2.167 − 1.100), yielding a net improvement in the overall team score.

To put this in context, the synergy advantage of 1.067 is comparable to the entire normalized quality score of a top-tier player such as Canyon (1.054), meaning the optimizer is willing to trade away most of a star's individual contribution to obtain it. This indicates that pairwise interactions

	Optimized roster		Greedy roster	
Top	Kiin	(1.437)	Kiin	(1.437)
Jungle	Canyon	(1.054)	Canyon	(1.054)
Mid	Chovy	(1.611)	Chovy	(1.611)
Bot	Aiming	(0.848)	Gumayusi	(0.988)
Support	Lehends	(0.735)	Delight	(1.304)
Total quality	5.685		6.393	
Total synergy	2.167		1.100	
Total score	7.852		7.494	

Table 2: Optimized vs. Greedy Rosters

can contribute a nontrivial share of overall team strength. It is also notable that the optimizer does not disperse the roster: it retains the highest-rated player at three positions and spends its budget rebuilding the bot side around players with shared history, rather than assembling five individually unremarkable players who happen to have played together.

6 Discussion and Impact

We have developed a robust linear integer optimization model to maximize overall team quality while explicitly accounting for player synergies. Using this approach, we identified teams that outperformed those constructed by simply selecting the top player for each role, potentially achieving higher performance at a lower cost. Importantly, the model is highly general and can be adapted to a wide range of team-based sports. The main challenge lies in accurately estimating player quality and synergy, which may require domain-specific analysis for each sport. With careful estimation, our approach provides a systematic way to assemble high-performing teams beyond simple heuristics.

7 Future Work

Currently, we estimate player synergy using observed win rates, defaulting to 50% when two players have not played together. This method introduces bias: part of the observed win rate reflects individual player quality rather than true synergy, and because rosters typically change infrequently, most player pairs have limited or no shared game history. Consequently, the available data is sparse. Future work could focus on developing more sophisticated estimators of synergy, possibly incorporating advanced statistical models or simulations to disentangle individual skill from interaction effects more effectively. This would lead to more accurate team evaluations and improved optimization outcomes.

References

[1] Esports Charts. LoL Esports Viewership and Statistics. `https://escharts.com/games/lol`. Accessed: August 27, 2026.

[2] Christine. LCK Awards 2024: Faker wins Player of the Year. `https://esports.gg/news/league-of-legends/lck-awards-2024-winners`, 2024. Accessed: August 27, 2026.

[3] John Popko. 2025 LCK Regular Season Voting Results. https://www.invenglobal.com/igec/articles/19644/2025-lck-regular-season-voting-results, 2025. Accessed: August 27, 2026.

[4] Tim Sevenhuysen. Oracle's Elixir: League of Legends esports stats. https://oracleselixir.com/, 2025. Accessed: August 27, 2026.